Adiabatic flash drum with binary liquid feed

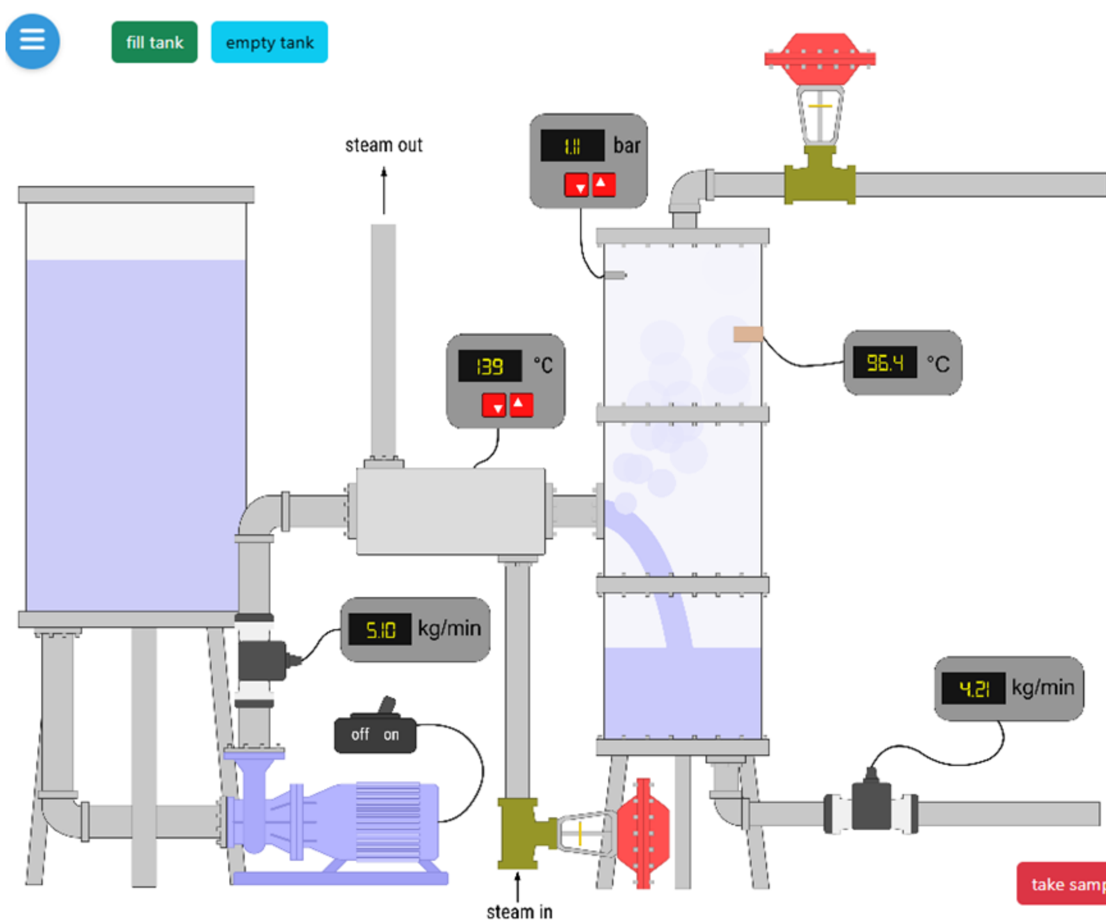
Name(s)____

A two-component, pressurized liquid partially vaporizes when it is fed to a lower-pressure adiabatic flash drum. Material balances are used to determine how much liquid vaporizes and the liquid and vapor compositions. A T-x_1-y_1 diagram is generated from measurements at constant drum pressure and different feed conditions, and the result is compared to Raoult's law.

Student learning objectives

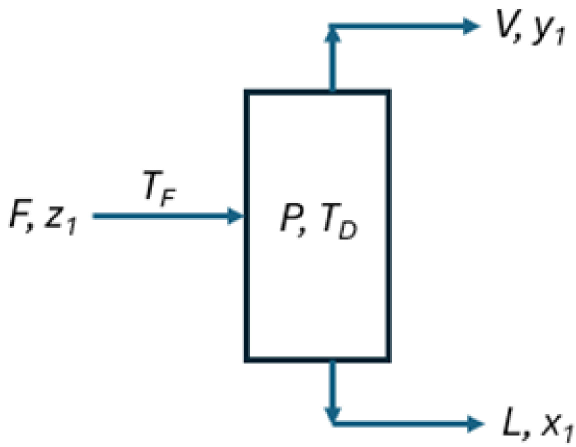
1. Explain the principles of flash distillation.
2. Do mass balances on a flash drum.
3. Create a T-x-y diagram for a binary mixture at constant pressure.
4. Apply Raoult's law.
5. Explain how changes in feed temperature and drum pressure affect the relative amounts of liquid and vapor, the compositions of the phases leaving the drum, and the separation ability of the drum.

Equipment



- A lab-scale insulated flash drum with liquid feed and vapor and liquid outlets. The drum contains pressure and temperature sensors.
- A system (with a temperature controller and temperature sensor) that preheats the feed mixture using steam.
- A pressure controller maintains constant pressure within the drum.
- Mass flow meters measure the flow rates of the feed and the liquid outlet.
- A gas chromatograph to analyze the composition of the liquid outlet.

Process flow diagram



Notation

z_i = mole fraction of component i in feed

y_i = mole fraction of component i in the vapor stream leaving the drum

x_i = mole fraction of component i in the liquid stream leaving the drum

T_F = feed temperature (^{o}C)

T_D = drum temperature (^{o}C)

P = drum absolute pressure (bar)

P_i^{sat} = saturation pressure of component i (bar)

MW_i = molecular weight of component i (g/mol)

$\overline{MW_L}$ = molecular weight of outlet liquid mixture (g/mol)

F = feed molar flow rate (60. mol/min)

L = outlet liquid molar flow rate (mol/min)

V = outlet vapor molar flow rate (mol/min)

$\dot{m}_F$ = feed mass flow rate (kg/min)

$\dot{m}_L$ = outlet liquid mass flow rate (kg/min)

$\dot{m}_V$ = outlet vapor mass flow rate (kg/min)

A_i, B_i, C_i = Antoine constants for component i

Questions to answer before starting the experiment

1. If feed temperature increases at constant pressure, will vapor flow rate increase or decrease? Explain.

2. Will the vapor be richer or leaner in the more volatile component?

3. If the drum pressure decreases at constant feed temperature, will more or less vapor form?

4. Why should vapor and liquid compositions be different?

Procedure for measurements at one condition

1. In the hamburger menu, select a feed mole fraction of chemical 1 in the binary mixture using the slider. Record in the Table below.
2. Select a binary mixture (A, B, C, or D) from the dropdown menu and record it in Table 1.
3. Record the Antoine constants for the two compounds in Table 1. Note that these constants are for an Antoine equation whose pressure is in mm Hg and temperature in ^{o}C. Convert the pressure to bar by multiplying it by 0.001333.
4. Record the molecular weights of the two components in Table 1.
5. Close the hamburger menu and click “fill tank.”
6. Click “on” to turn on the feed pump. The feed flow rate is 60. mol/min, and the feed pressure is high enough that the feed is liquid.
7. Increase the feed temperature by clicking on the red box with the triangle pointing up on the temperature sensor. The set temperature will display red until the feed temperature reaches the set temperature. It will then display white. The feed temperature needs to be high enough that vapor forms in the flash drum (also, the liquid flow rate will be less than the feed rate). Record the feed temperature T_F in Table 1. You may want to start a spreadsheet and record the values there.
8. Select the flash drum pressure by clicking on the red boxes on the pressure gauge. The pressure displays red until the selected pressure is reached. Record the drum pressure P in Table 1.
9. Allow the system to reach steady state.
10. Record the drum temperature T_D in Table 1.
11. Record the outlet liquid mass flow rate $\dot{m}_L$ in Table 1.

12. Click “take sample” of the liquid outlet stream and “inject sample” into the GC to measure its mole fraction. Record the mole fraction x_1 of component 1 in Table 1.
13. Calculate the molecular weight of the liquid mixture $\overline{MW}_L$ and record the value in Table 1.

$$\overline{MW}_L = x_1 MW_1 + x_2 MW_2$$

14. Use the liquid molecular weight to calculate the liquid molar flow rate L and record its value in Table 1. Remember that the mass flow rate is in kg/min and the molecular weight is in g/mol.

$$L = \frac{\dot{m}_L}{\overline{MW}}$$

15. Use mass balances to determine outlet vapor molar flow rate V and the mole fraction y_1 and record those values in Table 1.

$$F = L + V$$

$$F z_1 = L x_1 + V y_1$$

Table 1 Measurements for mixture ____

F = 60 mol/min P _________

A_1 _________ B_1 __________ C_1 ___________ MW_1 ___________

A_2 _________ B_2 __________ C_2 ___________ MW_2 ___________

T_F (°C)	z_1	T_D (°C)	$\dot{m}_L$ (kg/min)	x_1	$\overline{MW}_L$	L (mol/min)	V (mol/min)	y_1

Repeat these measurements; click “empty tank,” select a new feed mole fraction, and start a new experiment at the same drum pressure by repeating steps 5 to 15. To generate a T-x_1-y_1 diagram, make measurements to cover a feed composition range z_1 from 0.1 to 0.9, and adjust the feed temperature so that both vapor and liquid leave the drum. Record the results in Table 1.

Create a T-x-y diagram

Use the data in Table 1 to create a spreadsheet and generate a T-x_1-y_1 diagram at constant pressure.

Compare the experiment measurements to Raoult’s law by plotting Raoult’s law values on the same graph?

$$P = x_1 P_1^{sat} + (1 - x_1) P_2^{sat}$$

$$y_1 = \frac{x_1 P_1^{sat}}{P}$$

$P_i^{sat} = 10^{A_i - \frac{B_i}{T + C_i}}$ where T is in °C.

Questions to answer

1. Where might these measurements have errors?

2. In an adiabatic flash drum, why is the drum temperature lower than the feed temperature?

3. How does changing the feed temperature affect the outlet compositions and flow rates?

4. What happens if the feed temperature is too low?

5. What happens if the feed temperature is too high?

6. Suppose you repeat measurements at a lower drum pressure. How would the drum temperature change? How would the vapor flow rate change? How would the vapor composition change? Explain.

7. What safety precautions would you take to conduct this experiment in the laboratory?